

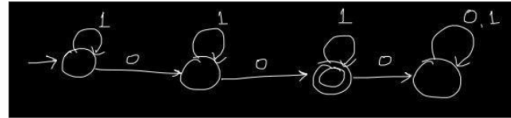
CSE331 Assignment 1 Solution
Fall 2025

Problem 1

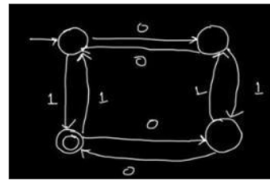
- a) $(0|1)^*(0|11|001)^*\epsilon|1|01$
- b) $abba(a|b|c)^*bac|abbac$
- c) $(0|1)(0|1)0(0|1)^*$
- d) $(1^*01^*01^*)^*|0^*10^*10^*$
- e) $(b^*ab^*ab^*ab^*)^*b^*ab^*$

Problem 2

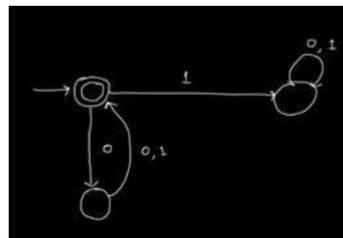
a)



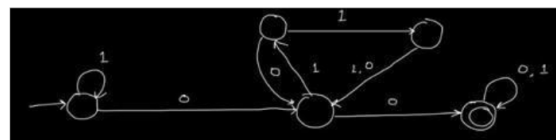
b)



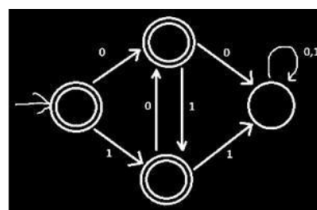
c)



d)



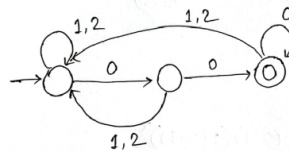
e)



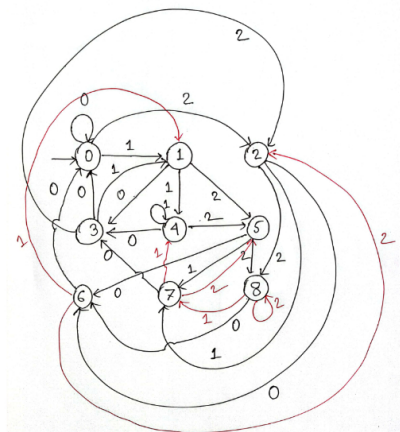
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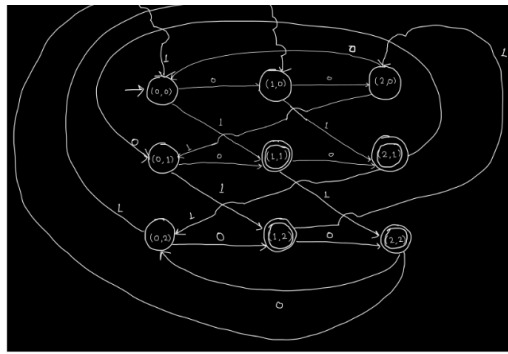
graph LR
    Start(( )) --> 00((00))
    00 -- 0 --> 01((01))
    00 -- 1 --> 10((10))
    01 -- 0 --> 01
    01 -- 1 --> 10
    10 -- 0 --> 01
    10 -- 1 --> 10
    style Start fill:none,stroke:none
    style 01 stroke-width:4px
    style 10 stroke-width:4px
  
```

In three base number system, the numbers which ends with 00. only those numbers are divisible by 9.



There is also a DFA with 9 states and 27 transitions - which is also correct. The logic is - if there is a number x , in ternary representation, adding a 0 at the end will make the number $3x$, adding 1 at the end will make it $3x+1$ and for 2 it will become $3x+2$.





A correct solution is anything equivalent to the above. The DFA diagram is a bit clunky due to there being a lot of states and transitions. However, the idea is simple: the states are indexed by a pair (i, j) where $0 \leq i, j \leq 2$, and i, j are the length and the number of 1s seen so far modulo 3. Upon seeing a new letter i goes up by 1, while j goes up by 1 only if the new letter is a 1.

The main observation is that $i^2 + j^2 = 2 \pmod 3$ iff $i = 1, 2$ and $j = 1, 2 \pmod 3$. This gives us four accepting states.

Any DFA for this language necessarily requires 9 states. So, any DFA with fewer states is wrong.